



On Estimating the Convective Mixing-Layer Height from Near-Surface Horizontal Wind Spectra: A Comment on Liu & Otaki (1997) with New Proposals

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Abstract

Estimating and monitoring the mixing-layer height remains a crucial but difficult task, as it cannot be measured directly by routine surface observations and is typically inferred from remote-sensing measurements with limited spatial coverage. In this context, the inversion of near-surface turbulence measurements from relatively simple, low-cost sonic anemometer offers a complementary solution. This paper reassesses the spectral approach proposed by Liu and Ohtaki in 1997 for determining the mixing-layer height, showing that its underlying assumptions are not physically justified, and presents a more consistent, although more complex, derivation. In addition, it presents two alternative methods, based on total and partial spectral integration, which yield comparatively simpler formulations.

Keywords Convective boundary layer · Mixing-layer height · Sonic anemometer · Turbulence spectra · Wind-velocity spectra

List of Symbols

z_i	Mixing-layer height
z	Measurement height above the ground
L	Obukhov length
n	Reduced frequency ($n = fz/\overline{U}$)
f	Frequency
$\overline{U}$	Mean wind speed at height z
$G_u(n)$	Normalized spectrum of the longitudinal velocity component
$G_v(n)$	Normalized spectrum of the transverse velocity component
$S_u(f), S_v(f)$	Power spectral densities of the horizontal wind components
u_*	Friction velocity
a_u, b_u, A_u, B_u	Empirical coefficients for the longitudinal component

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a_v, b_v, A_v, B_v	Empirical coefficients for the transverse component
$n_{u,max}$	Frequency corresponding to the maximum of G_u
$n_{v,max}$	Frequency corresponding to the maximum of G_v

1 Introduction

The mixing-layer height (z_i) is a characteristic length scale related to the vertical extent of turbulent motions within the atmospheric boundary layer (ABL), which mediates all exchanges of momentum, sensible and latent heat, trace gases, aerosols, and energy between the surface and the lower atmosphere. In this sense, z_i reflects the integrated contributions of mechanical and buoyant production of turbulence, entrainment at the top of the ABL, as well as the influence of large-scale subsidence and advection. As a result, z_i is not a geometric quantity, but rather a compact descriptor of the overall turbulent state of the ABL, playing a key role in virtually all numerical weather prediction, climate, and air-pollution models (Edwards et al. 2020).

Despite its importance, continuously determining and monitoring the mixing layer height remains challenging (e.g., Zhang et al. 2020). It cannot be measured directly by routine surface observations and is thus traditionally estimated from radiosoundings or, more frequently, from remote-sensing instruments such as LIDAR/ceilometers, microwave radiometers, wind profilers (see, e.g., Kotthaus et al. 2023 for an extensive review) and, especially under stable and very stable conditions, SODAR (Argentini et al. 2013; Petenko et al. 2014). The determination of z_i with each of these instruments is constrained by limited geographical coverage, depends on specific assumptions and captures only specific aspects of the ABL structure. While radiosounding profiles provide high-quality measurements but are spatially limited and suffer low temporal resolution, remote-sensing measurements are usually continuous in time but affected by instrument-specific biases, such as aerosol–turbulence mismatches for LIDAR, coarse vertical resolution for microwave radiometers, blind zones and weather sensitivity for wind profilers/RASS, as well as range and noise limitations for SODAR.

Among the approaches described in the literature, techniques based on turbulence measurements appear particularly suitable, as they are directly connected to the turbulent processes governing the ABL. Over the past decades, both prognostic (e.g., Batchvarova and Gryning 1994) and diagnostic (e.g., Casasanta et al. 2014) parametrizations were developed to estimate z_i under convective conditions using limited datasets, which is particularly valuable in situations where remote-sensing measurements are not available. In particular, building on the well-established spectral model proposed by Højstrup (1981), Liu and Ohtaki (1997) proposed a method to estimate z_i under convective conditions from horizontal wind velocity spectra, using surface turbulent parameters that can be obtained from relatively simple, low-cost sonic anemometers. While their approach appears promising, its performance critically depends on a correct treatment of both the low- and high-frequency spectral contributions, as well as on a robust and stable inversion procedure. Within this framework, our paper identifies critical flaws in the assumption of Liu and Ohtaki (1997), reassesses their method and proposes alternative strategies to retrieve z_i .

2 Theoretical Background

Building on previous experimental results of Kansan (1968) and Minnesota (1973) field campaigns (Kaimal et al. 1972; Kaimal 1978), and assuming a paradigmatic measurement site characterized by complete horizontal homogeneity, Højstrup (1981) proposed a spectral model for the three wind components u , v and w in the lower part of the unstable ABL. This model decomposes the spectral densities of the horizontal components $S_x(f)$ – with $x \in \{u, v\}$ – into low- and high-frequency contributions, expressed as a function of the reduced frequencies $n_i = fz_i/\bar{U}$ and $n = fz/\bar{U}$, respectively. Here, f denotes the frequency, z is the height from the ground and $\bar{U}$ is the mean wind speed at height z . Accordingly, the resulting equations for the spectra of the horizontal components are as follows:

$$G_u(n, n_i) = \frac{fS_u(f)}{u_*^2} = \frac{a_u n_i}{1 + b_u n_i^{5/3}} \left(\frac{z_i}{-L} \right)^{2/3} + \frac{A_u n}{(1 + B_u n)^{5/3}}, \quad (1a)$$

$$G_v(n, n_i) = \frac{fS_v(f)}{u_*^2} = \frac{a_v n_i}{(1 + b_v n_i)^{5/3}} \left(\frac{z_i}{-L} \right)^{2/3} + \frac{A_v n}{(1 + B_v n)^{5/3}}, \quad (1b)$$

where u_* is the friction velocity, L is the Obukhov length and the numerical coefficients $a_u, b_u, A_u, B_u, a_v, b_v, A_v, B_v$ are equal to 0.5, 2.2, 105, 33, 0.95, 2, 17 and 9.5, respectively.

Equations (1) include terms following either a blunt $Af/(1+Bf)^{5/3}$ or a pointed $Af/(1+Bf^{5/3})$ model. While the main characteristics of these two typical forms are summarized in Appendix 1 and extensively discussed in the literature (e.g., Olesen et al. 1984), it is worth highlighting that the asymmetry in the low-frequency terms – with the u -component following a pointed model and the v -component a blunt model – was introduced by Højstrup (1982) to better fit the observed differences between u - and v -spectra at low frequencies, while preserving inertial-subrange dissipation scaling.

Considering that the two reduced frequencies are not independent of each other, Eq. (1) can be rewritten in the following form:

$$G_u(n) = \left(\frac{z_i}{z} \right)^{5/3} \frac{a_u n}{1 + b_u \left(\frac{z_i}{z} n \right)^{5/3}} \left(\frac{z}{-L} \right)^{2/3} + \frac{A_u n}{(1 + B_u n)^{5/3}}, \quad (2a)$$

$$G_v(n) = \left(\frac{z_i}{z} \right)^{5/3} \frac{a_v n}{(1 + b_v \frac{z_i}{z} n)^{5/3}} \left(\frac{z}{-L} \right)^{2/3} + \frac{A_v n}{(1 + B_v n)^{5/3}}, \quad (2b)$$

The spectral model described above is a relatively simple two-range parametrization and, as such, does not aim to represent all the scaling regimes that emerge in more recent formulations, such as the Multipoint Monin–Obukhov Similarity (MMO, Tong and Nguyen 2015; Tong et al. 2026). Nevertheless, it remains widely used in the literature (see, e.g., Choi et al. 2011; Puccioni and Iungo 2021; Schiavon et al. 2023) and has been used in micrometeorology for several decades: for instance, it is implemented in the spectral corrections commonly applied in the Eddy-Covariance method to estimate surface-layer turbulence parameters (e.g., Aubinet et al. 2012). As is evident from the structure of Eqs. (1) or (2), their inversion would allow the estimation of z_i from the normalized spectra

of the horizontal wind components and from the prevailing turbulence level – i.e., from parameters commonly retrieved from low-cost sonic anemometers – albeit under a series of assumptions. First, measurements of the three wind components and of virtual potential temperature have to be available at a measurement height within the surface layer, and the experimental site has to be horizontally homogeneous and unaffected by non-turbulent perturbations such as waves or sub-mesoscale motions. The observation period T is assumed to be of the order of one hour or less, during which quasi-stationary conditions have to prevail. Then, the same dataset needs to allow the computation of the sampling spectra $G_u(n)$ and $G_v(n)$ over a frequency band extending from $1/T$ up to half the sampling frequency of the sonic anemometer, which is typically ≥ 10 Hz, with spectral resolution $1/T$ (Kaimal and Finnigan 1994). Finally, the absolute maxima of the two spectra ($f_{u,max}$ and $f_{v,max}$) have to be identified with sufficient accuracy, an operation that is not trivial given that these quantities tend not to be well defined (Panofsky and Dutton 1984).

To our knowledge, the only explicit attempt to invert Eqs. (1) was proposed by Liu and Ohtaki (1997), whose results have found limited applications, mostly related to the determination of mixing-layer scaling for profiles of potential-temperature and vertical-velocity variance (e.g., Barlow 2014 and references therein). Nevertheless, in the only validation attempt available in the literature, the methodology of Liu and Ohtaki performed poorly ($R^2 \approx 0.4$) when compared with estimates of z_i retrieved from tether-sonde observations (Choi et al. 2011). As will be discussed in the next section, this limited performance is not incidental, since the original formulation of their methodology contains a critical flaw that ultimately prevents its practical use.

3 Comments on the Method of Liu and Ohtaki (1997)

With reference to the spectral model illustrated in the previous Section (Eqs. 1), Liu and Ohtaki (1997) argue that, since the highest value of the spectral density for the horizontal wind components always occurs in the low-frequency range, the position of the spectral peak is “assisted” with n_i and can be determined by solving $\partial G_x / \partial n_i = 0$. For instance, when considering the longitudinal wind component u , the maximum of the low-frequency contribution occurs at $n_{i,max} \approx 0.795$, which, by definition, yields the following expression for the mixing-layer height:

$$z_i = 0.795 \frac{\bar{U}}{f_{max}}, \quad (3)$$

where f_{max} denotes the frequency at which the spectral peak is observed.

The previous equation is appealing for its simplicity but is intrinsically flawed, as its derivation completely neglects the high-frequency contribution associated with mechanical production of turbulence – so that the peak of G_x does not, in general, coincide with that of the low-frequency term alone – and implicitly treats n and n_i as independent variables, which is both physically and mathematically inconsistent.

In more detail, consider Eq. (2a) – which is equivalent to Eq. (1a) – as an example. In this case, the high-frequency contribution follows a blunt model, whose maximum value $G_{u,h,max} \approx 1.03$ always corresponds to $n_{h,max} \approx 0.045$. This contribution is thus not influenced by convection and depends only on z and $\bar{U}$ through n . Conversely, the low-frequency contribution is described by a pointed spectral form, and its peak frequency $n_{l,max}$ explicitly

depends on the ratio z/z_i according to $n_{l,max} = 0.795z/z_i$. As a result, the corresponding maximum value depends on both the mixing-layer height and the level of thermal turbulence, $G_{u,l,max} = 0.1589 (z_i/L)^{2/3}$.

To evaluate how the position of the maximum of $G_u(n)$ varies as a function of z_i and L , a numerical experiment was performed using realistic values of both z/z_i and $-z/L$ under convective conditions (Stull 1988), spanning the ranges 0.003–0.05 and 0.01–1, respectively. The resulting spectra, presented in Appendix 2, were computed over a reduced frequency range $n \in [10^{-4}, 10^2]$, with a resolution of 10^{-4} , thus covering six orders of magnitudes. Despite such a fine resolution, the determination of n_{max} , corresponding to the absolute maximum of $G_u(n)$, remains affected by a non-negligible uncertainty. The results are summarized in Table 1, which lists $n_{h,max}$ and $n_{l,max}$ as a function of four values of z/z_i , together with the corresponding n_{max} values obtained for five levels of convection expressed in terms of $-z/L$. As expected, n_{max} ranges within roughly one order of magnitude between $n_{l,max}$ and $n_{h,max}$. For $z/z_i = 0.003$, n_{max} remains close to $n_{h,max}$ up to $-z/L = 0.05$, and then decreases by approximately one order of magnitude towards $n_{l,max}$ as mechanical production of turbulence is gradually overtaken by the convective contribution. As z/z_i increases, this transition requires progressively higher values of $-z/L$, and for $z/z_i = 0.05$ the spectral maximum remains essentially controlled by the mechanical contribution. Similar considerations apply when the same analysis is carried out starting from Eq. (2b), indicating that, at least under weak-to-moderate convective conditions, the position of the spectral maximum remains close to $n_{h,max}$, in clear contrast with the assumption underlying Eq. (3).

A more consistent expression for estimating the mixing-layer height from Eqs. (2) can be derived by following the procedure adopted by Liu and Ohtaki (1997), but without neglecting the high-frequency contribution, yielding the following expressions:

$$\frac{z_i}{z} = \left(\frac{z}{-L} \right)^{-2/5} \left[\frac{3 \left(1 + b_u \left(\frac{z_i}{z} \right)^{5/3} n_{u,max}^{5/3} \right)^2}{a_u \left(3 - 2b_u \left(\frac{z_i}{z} \right)^{5/3} n_{u,max}^{5/3} \right)} C_u \right]^{3/5}, \quad (4a)$$

$$\frac{z_i}{z} = \left(\frac{z}{-L} \right)^{-2/5} \left[\frac{\left(1 + b_v \left(\frac{z_i}{z} \right) n_{v,max} \right)^{8/3}}{a_v \left(3 - 2b_v \left(\frac{z_i}{z} \right) n_{v,max} \right)} C_v \right]^{3/5}, \quad (4b)$$

Table 1 Normalized frequencies corresponding to the high-frequency (mechanical) peak $n_{h,max}$, the low-frequency (convective) peak $n_{l,max}$, and the absolute spectral maximum $n_{u,max}$ of $G_u(n)$, reported as a function of z/z_i and for different values of $-z/L$

z/z_i	$n_{h,max}$	$n_{l,max}$	$n_{u,max}$				
			$-z/L$				
			0.01	0.05	0.1	0.5	1
0.003	0.04545	0.00238	0.0394	0.0260	0.003	0.0025	0.0025
0.006		0.00477	0.0397	0.0280	0.0127	0.0057	0.0053
0.01		0.00795	0.0402	0.0305	0.0226	0.0107	0.0095
0.05		0.03973	0.0450	0.0443	0.0438	0.0423	0.0417

with

$$C_u = -\frac{A_u(3 - 2B_u n_{u,max})}{3(1 + B_u n_{u,max})^{8/3}},$$

$$C_v = -\frac{A_v(3 - 2B_v n_{v,max})}{3(1 + B_v n_{v,max})^{8/3}}.$$

Clearly, Eq. (4) are considerably more complex than the simple formula provided by (3) and do not admit a closed-form for solution, but can be solved numerically—for instance, by the fixed-point iteration method (Conte and De Boor 1980).

4 Alternative Approaches for inverting the Højstrup Model

This section presents two alternative approaches aimed at overcoming the methodological flaws in Liu and Ohtaki (1997) discussed in the previous Section. While theoretically consistent, their practical applicability and reliability still need to be assessed through dedicated experimental validation, which is also necessary to quantify the uncertainty associated with the resulting estimates of z_i . The Højstrup model is based on the decomposition of the horizontal velocity spectra into two distinct contributions: a mechanical, high-frequency component, which is always present, and a convective, low-frequency component, which vanishes under adiabatic conditions and increases with both thermal turbulence and mixing-layer height. While formally simple, such a formulation presents critical aspects. In particular, the high-frequency mechanical contribution may be overestimated under certain conditions (Lothion et al. 2009) and, more importantly, the numerical coefficients of the model were derived from a limited set of historical experiments and are provided without associated uncertainty estimates. In the following, these coefficients are assumed to be exact, while the potential limitations arising from this assumption are explicitly highlighted.

4.1 Total-Variance-Based Inversion Method

The total-variance method exploits the relationship between the spectral density and the variance of the turbulent velocity components, namely,

$$\sigma_u^2 = \int_0^\infty S_u(f) df. \quad (5)$$

Substituting Eq. (2a) in (5) yields (see Appendix 1):

$$\frac{\sigma_u^2}{u_*^2} = C_{u,l} + C_{u,h}, \quad (6)$$

With

$$C_{u,l} = \frac{1.98a_u}{b_u^{\frac{3}{5}}} \left(\frac{z_i}{-L} \right)^{\frac{2}{3}}, C_{u,h} = \frac{3}{2} \frac{A_u}{B_u}.$$

Similarly, the expression for the variance of the transverse component is retrieved by inserting Eq. (2b), rather than Eq. (2a), into Eq. (5). It is worth highlighting that, aside from the differences in the numerical values of the coefficients, the previous equation is formally identical to the expression originally postulated by Wyngaard and Coté (1974) and later refined in Panofsky et al. (1977). From Eq. (6), the mixing-layer height can then be directly obtained as,

$$z_i = z \left[\frac{\sigma_u^2/u_*^2 - C_{u,h}}{C_{u,l}/(-L)^{2/3}} \right]^{3/2}, \quad (7)$$

where $C_{u,l} \approx 0.62$ and $C_{u,h} \approx 4.8$.

The previous equation provides an explicit estimation of z_i without requiring numerical methods, but it is sensitive to uncertainty and potential contaminations in the estimates of σ_u^2 and u_* , particularly under weakly convective conditions, which may cause the numerator to become unreliably small or even negative.

4.2 Partial-Spectrum Integration Method

Another approach to estimate the mixing-layer height by inverting Eqs. (2), partly inspired by Højstrup et al. (1997) and Dias (2018), consists of performing a partial integration of the velocity spectrum over the frequency band from 0 to f_{max} , which represents approximately half of the total variance (Højstrup et al. 1997). With reference to Appendix 1, applying this partial integration to Eq. (2b), for instance, leads to the following expression,

$$\frac{\sigma_{max}^2}{u_*^2} = \left(\frac{z_i}{z} \right)^{2/3} \left(\frac{z}{-L} \right)^{2/3} \frac{3a_v}{2b_v} \left[1 - \frac{1}{\left(1 + b_v \left(\frac{z_i}{z} \right) n_{max} \right)^{2/3}} \right] + \frac{3A_v}{2B_v} \left[1 - \frac{1}{\left(1 + B_v n_{max} \right)^{2/3}} \right], \quad (8)$$

which can be rewritten as:

$$\frac{z_i}{z} = C_1^{\frac{3}{2}} \left[1 - \frac{1}{\left(1 + b_v \left(\frac{z_i}{z} \right) n_{max} \right)^{2/3}} \right]^{-\frac{3}{2}}, \quad (9)$$

With

$$C_1 = \frac{2b_v}{3a_v} \left(\frac{z}{-L} \right)^{-2/3} \left(\frac{\sigma_{max}^2}{u_*^2} - \frac{3A}{2B} \left[1 - \frac{1}{\left(1 + B n_{max} \right)^{2/3}} \right] \right).$$

As with Eq. (4), Eq. (9) does not admit a closed-form solution and has to be solved numerically.

Applying the same method to the longitudinal component u (Eq. 2a) is slightly more complicated, since in this case the low-frequency contribution follows a pointed model and its analytical integration is not trivial. To overcome this issue, it is convenient to replace the original pointed spectral form with an equivalent blunt form, specifically chosen to preserve both the variance and the frequency at which the spectral maximum occurs (Appendix 1). Equation (2a) becomes:

$$G_u(n) = \left(\frac{z_i}{z}\right)^{5/3} \frac{\alpha n}{\left(1 + \beta\left(\frac{z_i}{z}\right)n\right)^{5/3}} \left(\frac{z}{-L}\right)^{2/3} + \frac{A_u n}{(1 + B_u n)^{5/3}}, \quad (10)$$

with $\alpha = 1.552 a_u$ and $\beta = \frac{3}{2} \left(\frac{2}{3} b_u\right)^{3/5}$. With this approximation, the method described above can be applied to the longitudinal component as well.

5 Conclusions

The analysis presented in this work revisits the method proposed by Liu and Ohtaki (1997) for estimating the mixing-layer height from horizontal velocity spectra, showing that its original formulation contains critical flaws. Although nearly three decades have passed since its publication, such a reassessment is still relevant, as the possibility of retrieving z_i from near-surface turbulence measurements is becoming increasingly relevant thanks to the widespread availability of modern sonic – anemometers.

It was demonstrated that the assumption underlying the Liu and Ohtaki approach – namely, the omission of the high-frequency mechanical term and the inconsistent treatment of normalized frequencies – cannot be justified on physical grounds. When the inversion procedure is carried out correctly, the resulting expressions for z_i are more complicated than the simple formula presented in 1997 and do not admit a closed-form solution; however, they are fully consistent with the structure of the Højstrup (1981) spectral model and the expected behaviour of the convective mixing layer.

In addition, two alternative inversion strategies were proposed, based on partial and total integration of the spectrum, which lead to comparatively simpler formulations. Nevertheless, it should be emphasized that the methods presented in this paper have not yet been experimentally assessed, and their performance in real applications is expected to depend on the reliability of the numerical coefficients in the Højstrup model and on their associated uncertainties, which is not adequately addressed in the literature. Thus, the proposed inversion methods should be regarded as a conceptual framework rather than a ready-to-use operational tool. Future work should address the validation of these methods using field turbulence datasets and assess both their operational feasibility and their sensitivity to uncertainties. At the same time, novel approaches may arise from the adoption of more recent models (Tong and Nguyen 2015), that may provide a more complete physical representation of the spectrum.

Appendix 1: Analytical Properties of the Blunt and Pointed models

Measurements collected during the 1968 Kansas experiment and the 1973 Minnesota experiment (Kaimal et al. 1972; Kaimal 1978) suggest that the spectra of the three wind components are well represented by two spectral models—or by a linear combination of them—whose main analytical properties are described below (Olesen et al. 1984; Sorbjan 1989).

Blunt Model

The blunt form is used to model the spectra of the horizontal wind components, the potential temperature and the concentration of passive scalars under convective conditions. For a generic variable x , it is defined as,

$$G_{xx}^b(f) = fS_{xx}(f) = \frac{Af}{(1 + Bf)^{5/3}}, \quad (11)$$

and its asymptotic behaviour,

$$G_{xx}^b(f) \Big|^{Asym} = \frac{A}{B^{5/3}} f^{-2/3}, \quad (12)$$

is fully consistent with Kolmogorov's 1941 theory (K41). The frequency at which Eq. (11) reaches a maximum is,

$$f_{max} = \frac{3}{2B}, \quad (13)$$

which corresponds to a $G_{xx}^b(f)$ value of,

$$G_{max}^b = 0.326 \frac{A}{B}. \quad (14)$$

In addition, since the integral of the spectrum $S_{xx}(f)$ between 0 and a generic frequency f is,

$$\int_0^f S_{xx}(f') df' = \frac{3}{2} \frac{A}{B} \left[1 - \frac{1}{(1 + Bf)^{2/3}} \right], \quad (15)$$

the variance of x given by,

$$\sigma_x^2 = \int_0^\infty S_{xx}(f) df = \frac{3}{2} \frac{A}{B}. \quad (16)$$

Pointed Model

The pointed form is appropriate for representing all variables under stable conditions, and the vertical wind component under both stable and convective conditions.

For a generic variable x , it is defined as,

$$G_{xx}^p(f) = fS_{xx}(f) = \frac{af}{1 + bf^{5/3}}, \quad (17)$$

and, as in the case of the blunt model, its asymptotic behaviour:

$$G_{xx}^p(f)|^{Asym} = \frac{a}{b}f^{-2/3} \quad (18)$$

is fully consistent with Kolmogorov's 1941 theory (K41). The frequency corresponding to the maximum of Eq. (17) is,

$$f_{max} = \left(\frac{3}{2b}\right)^{\frac{3}{5}}, \quad (19)$$

for which the corresponding value of $G_{xx}^b(f)$ is,

$$G_{max}^p = 0.510 \frac{a}{b^{3/5}}. \quad (20)$$

The integral of the spectrum from zero to infinity, corresponding to the variance of x , is given by (Olesen et al. 1984):

$$\sigma_x^2 = \frac{3}{5} \frac{a}{b^{3/5}} \frac{\Gamma(3/5)\Gamma(2/5)}{b} = 1.98 \frac{a}{b^{3/5}}. \quad (21)$$

Unfortunately, in this case deriving an expression for the integral of the spectrum from zero to a generic frequency f is not straightforward, and it is more convenient to replace the pointed form with an equivalent blunt form that preserves both the total variance and the frequency at which the spectrum peaks:

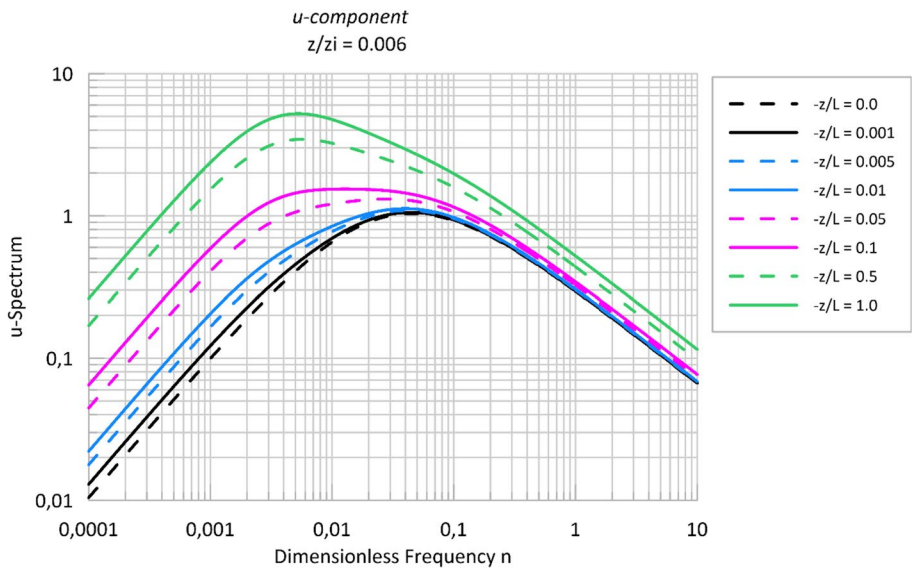
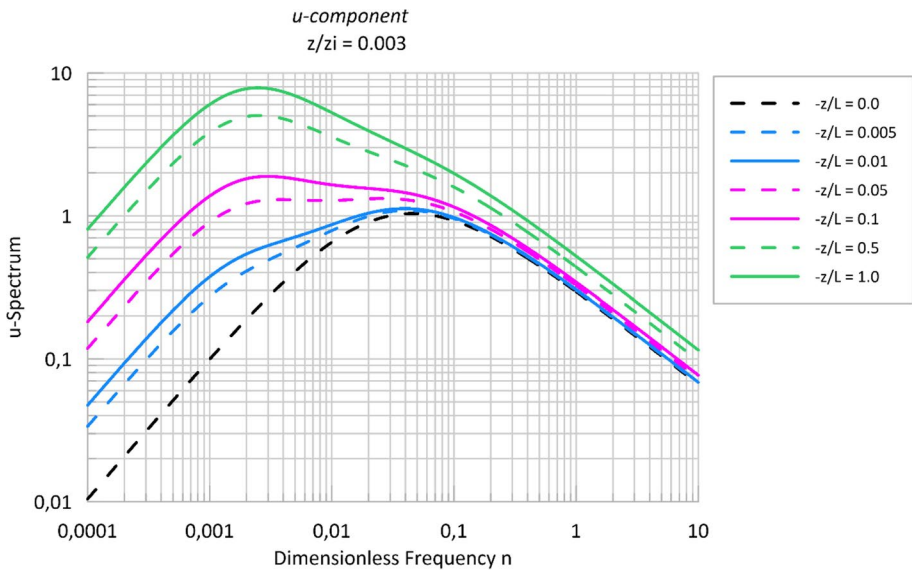
$$G_{xx}^{p,e}(f) = \frac{\alpha f}{(1 + \beta f)^{5/3}}, \quad (22)$$

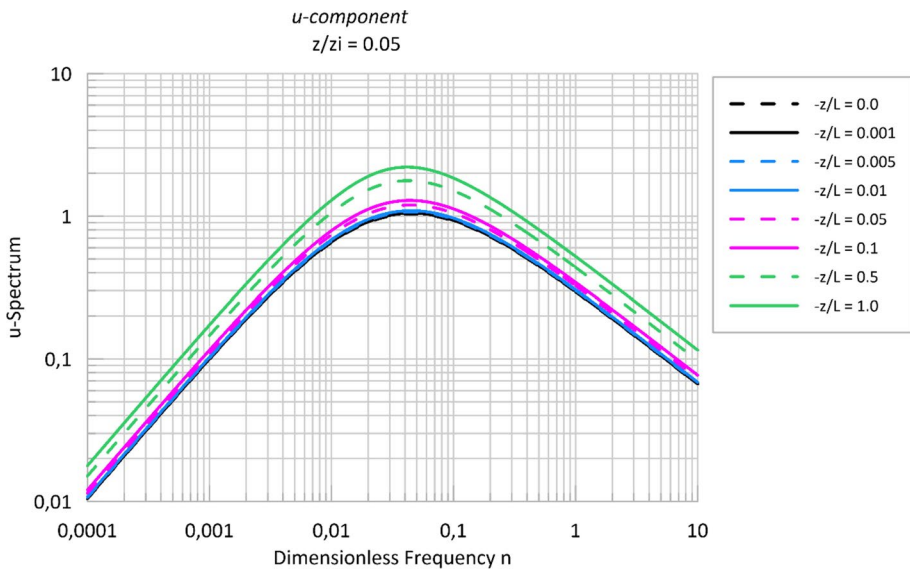
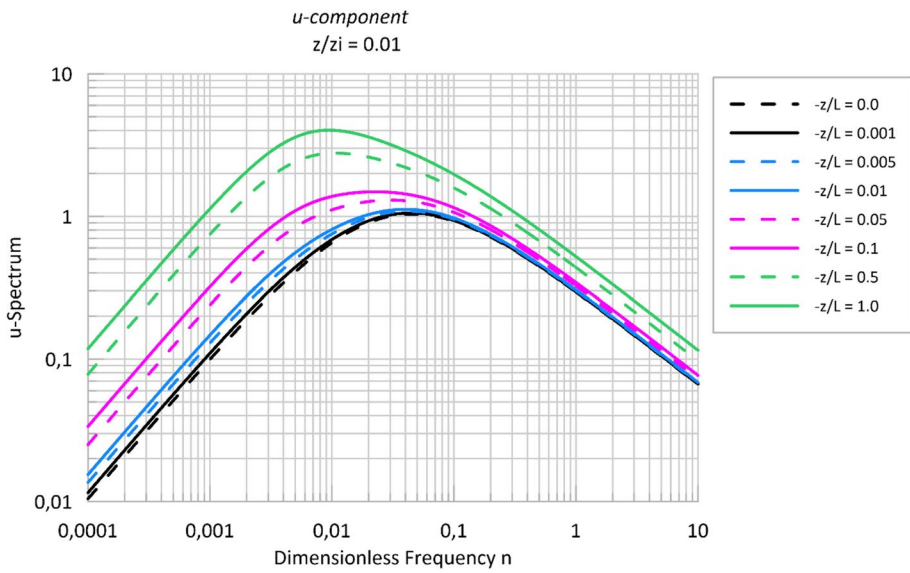
With

$$\alpha = 1.55a \quad \beta = \frac{3}{2} \left(\frac{2b}{3}\right)^{3/5}.$$

Appendix 2: Numerical Reconstruction of $G_u(n)$

Numerically reconstructed spectra for the longitudinal (u) velocity component, performed for $z/z_i = 0.003, 0.006, 0.01, 0.05$ and for $-z/L = 0.0, 0.001, 0.005, 0.01, 0.05, 0.1, 0.5, 1$.





Appendix 3: Derivation of Eqs. (4a) and (4b)

To clarify the derivation of Eqs. (4), we briefly outline the inversion procedure of the spectral model given in Eqs. (2). Let us consider Eq. (2a) for the longitudinal velocity component:

$$G_u(n) = \left(\frac{z_i}{z}\right)^{5/3} \frac{a_u n}{1 + b_u \left(\frac{z_i}{z} n\right)^{5/3}} \left(\frac{z}{-L}\right)^{2/3} + \frac{A_u n}{(1 + B_u n)^{5/3}}, \quad (2a)$$

The position of the spectral maximum is obtained by imposing:

$$\frac{dG_u}{dn} = 0. \quad (23)$$

Differentiating Eq. (2a) with respect to n , while treating z_i/z as a constant, yields:

$$a_u \left(\frac{z_i}{z}\right)^{5/3} \left(\frac{z}{-L}\right)^{2/3} \frac{1 - \frac{2}{3} b_u \left(\frac{z_i}{z} n\right)^{5/3}}{\left[1 + b_u \left(\frac{z_i}{z} n\right)^{5/3}\right]^2} + A_u \frac{1 - \frac{2}{3} B_u n}{(1 + B_u n)^{8/3}} = 0, \quad (24)$$

evaluated at $n = n_{u,max}$.

Rearranging the above expression, one obtains:

$$\left(\frac{z_i}{z}\right)^{5/3} = -\frac{A_u}{a_u} \left(\frac{z}{-L}\right)^{-2/3} \frac{1 - \frac{2}{3} B_u n_{u,max}}{(1 + B_u n_{u,max})^{8/3}} \frac{\left[1 + b_u \left(\frac{z_i}{z} n_{u,max}\right)^{5/3}\right]^2}{1 - \frac{2}{3} b_u \left(\frac{z_i}{z} n_{u,max}\right)^{5/3}}. \quad (25)$$

Finally, noting that z_i appears on both sides of the equation, the above expression corresponds to Eq. (4a) and must be solved numerically.

An analogous procedure can be applied to Eq. (2b) for the transverse velocity component, leading to Eq. (4b). By differentiating Eq. (2b), one obtains:

$$a_v \left(\frac{z_i}{z}\right)^{5/3} \left(\frac{z}{-L}\right)^{2/3} \frac{1 - \frac{2}{3} b_v \left(\frac{z_i}{z} n\right)}{\left[1 + b_v \left(\frac{z_i}{z} n\right)\right]^{8/3}} + A_v \frac{1 - \frac{2}{3} B_v n}{(1 + B_v n)^{8/3}} = 0. \quad (26)$$

evaluated at $n = n_{v,max}$, which leads to Eq. (4b). As for the longitudinal component, the resulting expression is implicit in z_i and must be solved numerically.

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Data Availability The datasets analyzed during the current study are not publicly available but are available from the corresponding author upon reasonable request.

Declarations

Conflict of interest The authors declare no competing interests.

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